

Soma LingYa

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Soma LingYa

Soma LingYa: Parameter-Efficient Fine-Tuning via LingYa Channel

Abstract

1. Introduction

1.1

1.2 LoRA

$$\Delta W = B \cdot A \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$$

$$|\Theta_{\text{LoRA}}| = 2 \cdot d \cdot r$$

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1.4

2. Method

2.1

$$\Delta W = R \cdot P \cdot \alpha$$

$$W' = W + \Delta W = W + R \cdot P \cdot \alpha$$

2.2 R

2.3 Delta Clamp

$$\text{if } |P|_F > \tau_{\max}: \quad P \leftarrow P \cdot \frac{\tau_{\max}}{|P|_F}$$

2.4

$$W_{\text{fused}} = W_{\text{orig}} + R \cdot P \cdot \alpha$$

3. Experiments

3.1

3.2

$2dr$	dr			
512	4	4,096	2,048	**50.0%**
512	8	8,192	4,096	**50.0%**
512	16	16,384	8,192	**50.0%**

3.3 Delta Clamp

3.4

3.5

3.6

PPL			
ROOT	2,048	23.1	600
ROOT + 2×BRANCH	4,096	22.8	500
ROOT + 2×BRANCH + LEAF	6,144	**22.5**	400
BRANCH	4,096	22.9	550

3.7

$\tau_{\max}$	PPL	
1.0	23.5	
5.0	22.8	
10.0	23.2	

4. Discussion

4.1 LoRA

LoRA	Soma LingYa	
	$\Delta W = B \cdot A$	$\Delta W = R \cdot P$
	$2dr$	dr
	✓	✓
		×

4.2 1

$$\|R \cdot P - \Delta W_{\text{true}}\|_F \leq \|\Delta W_{\text{true}}\|_F \cdot \sqrt{1 - \frac{r}{d_{\text{out}}}}$$

4.3

4.4

5. Conclusion

References

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